

Victoria Nguyen

682-407-4120 | victoria.n.nguyen05@gmail.com | [linkedin.com/in/vicngan](https://www.linkedin.com/in/vicngan) | github.com/vicngan | vicnganfolio.netlify.app

EDUCATION

University of Michigan - Ann Arbor

Bachelor of Science in Computer Science, Minor in Entrepreneurship

Ann Arbor, MI

May 2028

- **Relevant Coursework:** Data Structures and Algorithms, Discrete Math, Entrepreneurship Practicum, Digital Product Design

WORK EXPERIENCE

Gametime United

Incoming Full-Stack Software Development Intern

Remote

June 2026

Michigan Medicine - Department of Computational Medicine

Research Assistant – Genomic Agent Platform Development

Ann Arbor, MI

Jan 2026 – Present

- **Defined product requirements** for a conversational AI interface enabling scientists to query DNA binding predictions across the **3B+ base-pair human genome**, translating complex ML model outputs into intuitive user interactions.
- **Architected a FastAPI middleware layer** connecting React interfaces with transformer-based genomic models, reducing query latency for 50kb genomic windows by **40%** through caching optimization and asynchronous I/O.
- **Partnered with PhD researchers** to convert ML outputs into interpretable product features – including motif highlight maps and visualization layers – bridging technical capabilities with end-user needs.
- **Drove iterative feature development** through structured researcher feedback cycles, improving usability and streamlining genomic data exploration workflows.

Michigan Medicine - Department of Pediatrics

Research Assistant – Data Analysis & Behavioral Research

Ann Arbor, MI

Apr 2025 – Feb 2026

- **Built Python data pipelines** to clean and standardize large behavioral datasets, integrating validation logic that eliminated data loss and recovered **15+ hours of manual work per week**.
- **Achieved 90%+ inter-rater reliability** in Observer XT behavioral coding, producing a high-quality labeled dataset used as ground truth for downstream statistical analysis and ML research.
- **Analyzed 60+ pediatric feeding interaction recordings** to extract structured behavioral metrics, directly informing caregiver-response models and research prioritization decisions.

PROJECTS

SynthLab | React, FastAPI, Python, FAISS, SDV, RAG

Dec 2025 – Feb 2026

- **Led full-stack development** of an AI research platform supporting secure synthetic data generation for clinical trial simulations, owning the product from architecture through researcher-facing features.
- **Designed a synthesis pipeline** enforcing physiological constraints to generate medically plausible datasets, ensuring outputs met domain standards for simulation and downstream modeling use cases.
- **Built a semantic literature intelligence engine** using FAISS vector embeddings and LLM APIs, enabling natural-language querying and automated summarization across large research PDF libraries.

Heartline Care Companion | Python, Streamlit, React Native, SQLite

Oct 2025 – Nov 2025

- **Designed a cross-platform digital health application** supporting mood and energy tracking with caregiver communication features, applying user-centered thinking to a sensitive clinical context.
- **Built analytics dashboards** using Matplotlib and Altair to surface longitudinal health trends across **30-day windows**, translating raw data into actionable insights for caregivers.
- **Implemented automated alert workflows** via Twilio SMS and SMTP, triggering caregiver notifications when symptom metrics deviated **20% from patient baselines**.

PRODUCT & DESIGN PROJECTS

BetterSchedule | Figma, User Research, Design Thinking, Agile Methodologies

- Applied **design thinking** to develop a scheduling product concept, leading user research and usability testing to identify pain points and define feature priorities from interview insights.
- Translated user research findings into **wireframes, journey maps, and interactive Figma prototypes**, iterating through critique cycles to validate product decisions before development.
- **Collaborated cross-functionally with engineering and design peers** to align on product requirements, scope features against user needs, and move concepts from research insights to testable prototypes.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (ES6+), TypeScript, SQL, HTML/CSS, Bash

Frameworks: React, FastAPI, Node.js, Express, React Native, TailwindCSS

Data & AI: Pandas, NumPy, FAISS Vector Search, Synthetic Data Vault, RAG, Plotly, D3.js, LLM APIs/SDK

Product & Design: Agile/Scrum, Sprint Planning, User Research, Design Thinking, Figma, Canva, UI/UX Prototyping

Tools & Infrastructure: Git/GitHub, Docker, MongoDB, REST APIs, Google Cloud, Jupyter, Adobe Express